

Young Hyun Kim

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Dept. of Integrative Bioscience & Technology, Sejong University, Seoul, Republic of Korea

RESEARCH INTERESTS

Human cardiac tissue engineering, chamber-specific electrophysiology, mechanobiology, and quantitative electromechanical phenotyping using hiPSC-derived models.

EDUCATION

Sejong University

Seoul, Republic of Korea

M.S. in Integrative Biological Sciences and Industry

Aug 2026

Thesis: Generation of Chamber-Specific 3D Cardiac Organoids and an Integrated Electro-Mechanical Optical Analysis Platform.

B.S. in Integrative Biological Sciences and Industry

Aug 2023

RESEARCH EXPERIENCE

Biological Systems Engineering Laboratory, Sejong University

Seoul, Korea

Dept. of Integrative Bioscience & Technology · PI: Prof. Keel Yong Lee

Graduate Research Assistant

Sep 2023 – Present

Undergraduate Research Assistant

Oct 2022 – Aug 2023

Thesis project — Chamber-specific cardiac organoid platform

- **Generation:** Generated chamber-specific atrial- and ventricular-like 3D cardiac organoids via temporal Wnt/retinoic-acid modulation.
- **Structural phenotyping:** Characterized chamber identity by qRT-PCR, immunofluorescence, and flow cytometry; developed a MATLAB-based morphology analyzer validated against manual measurements.
- **Functional phenotyping:** Developed and validated iCAMAnalyzer, a Python platform integrating calcium kinetics, activation/conduction mapping, and label-free contractility; achieved ~7% error against simulated conduction data and enabled automated 9-organoid analysis.

Additional Projects & Lab Responsibilities

- Developing a spinner-flask protocol to scale hiPSC-cardiomyocyte production for higher-throughput organoid fabrication.
- Engineered C2C12-based 3D skeletal muscle tissues on custom micropillar scaffolds informed by COMSOL simulations; assessed contractility and calcium-signal propagation.
- **Collaborative contribution:** Supplied hiPSC-derived cardiomyocytes and contributed Resources/Visualization to a co-authored Cancer Research Communications study.

Undergraduate Research Highlights

- Developed the lab's initial iPSC-derived cardiac organoid fabrication protocol, establishing a foundation for subsequent organoid work.
- Fabricated C2C12-based 3D skeletal muscle tissues and applied cyclic mechanical stretch to assess tissue maturation.
- **Founding laboratory member:** As one of three core members, contributed to establishing the laboratory's initial infrastructure, including equipment setup, core experimental SOPs, and safety procedures.

PUBLICATIONS

1. **Kim, Y. H.**, Son, Y. H., Choi, Y., Kim, M. S., Park, S. J., & Lee, K. Y. (2026). Advanced in vitro cardiac models for drug evaluation: Integration of organoids, engineered tissues, and microphysiological systems. *Microsystems & Nanoengineering*, 12(1), 162. **(Review; First author)**
2. Kim, M. S., **Kim, Y. H.**, Choi, Y., & Lee, K. Y. (2025). Biohybrid actuators in robotics: recent trends and future perspectives of skeletal and cardiac muscle integration. *npj Robotics*, 3(1), 37. (Review; Second author)
3. Kim, Y. S., Lee, M., Cheon, Y. J., Choi, H. S., Choi, B. J., **Kim, Y. H.**, Lee, K. Y., & Ryu, C. J. (2026). Targeting CD147 or CD98hc is sufficient to collapse the CD147–CD98hc–MCT1/MCT4 module and restrain hepatocellular carcinoma growth. *Cancer Research Communications*. (Contributed Resources, Visualization)

CONFERENCE PRESENTATIONS

- Poster, Korea BioChip Society (Nov 2025, Jeju) — Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis
- Poster, 2025 Summer Symposium on Biological Convergence Engineering (Sep 2025, Sejong University) — Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

INSTRUMENTATION & ENGINEERING PROJECTS

- **Bio-Synchronous 3D Tissue Electromechanical Stimulator (2025–present, in progress):** Leading the functional redesign of an existing stretch system to integrate mechanical stretch, electrical stimulation, and controlled culture conditions; defining experimental requirements and coordinating implementation with an external fabricator.

TEACHING AND MENTORING — Sejong University

Undergraduate Research Mentor — SURF Program Jul 2026 – Aug 2026

Mentored 4 undergraduate researchers in laboratory onboarding, C2C12 and hiPSC culture, experimental planning, and preparation of final project presentations.

Teaching Assistant — Biochemistry Laboratory Jul 2023 – Dec 2024

Guided ~120 students through antibody-production workflows including plasmid transformation, protein expression and purification, and Western blotting; prepared instructional slides and laboratory materials.

Teaching Assistant — Microbiology Laboratory Jan 2024 – Jun 2024

Instructed 40 students in bacterial culture, Gram staining, morphological characterization, biochemical testing, and API-based bacterial identification.

Peer Research Mentor — DoLearn-DoRun Program Sep 2023 – Dec 2023

Mentored 2 undergraduate researchers on an in vitro muscle model using line-patterned substrates to promote myotube alignment; authored the project proposal and directed experimental work.

SKILLS

Cell & Tissue Engineering: hiPSC culture and chamber-specific cardiomyocyte differentiation; 3D cardiac organoid and skeletal muscle tissue engineering.

Assays & Imaging: flow cytometry, qRT-PCR, immunofluorescence, calcium-transient imaging, and contractility analysis.

Computational & Programming: MATLAB (App Designer), Python (Streamlit, NumPy/SciPy), ImageJ, GraphPad Prism; custom analysis-tool development.

Simulation, Fabrication & Illustration: COMSOL Multiphysics; Rhino/3D printing; Inkscape/Illustrator.